Question 1

Launch the Apache web server service using terminal commands.

Approach

Use service manager to start Apache and verify it is running.

Commands

- `systemctl` → manage linux services
- `service` → start/stop services
- `status` → verify service state

Output

STEP 1 : start apache service

```
┌──(aman㉿kali)-[~]
└─$ sudo systemctl start apache2
[sudo] password for aman:
```

STEP 2 : verify service status

```
┌──(aman㉿kali)-[~]
└─$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
     Active: active (running) since Fri 2026-04-03 05:46:46 IST; 16s ago
 Invocation: 74be5fc0dec645fe8ce446f5098295de
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 1765 (apache2)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 6 (limit: 9288)
     Memory: 21.2M (peak: 21.4M)
        CPU: 241ms
     CGroup: /system.slice/apache2.service
             ├─1765 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─1789 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─1791 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─1792 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─1793 /usr/sbin/apache2 -k start -DFOREGROUND
             └─1794 /usr/sbin/apache2 -k start -DFOREGROUND

Apr 03 05:46:45 kali systemd[1]: Starting apache2.service - The Apache HTTP Server...
Apr 03 05:46:46 kali apachectl[1765]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
Apr 03 05:46:46 kali systemd[1]: Started apache2.service - The Apache HTTP Server.
```

Key Observations

Apache runs on port 80 by default.
Service must be active to host webpages.

What I learned

Web servers are controlled using system services.

Question 2

Modify the default webpage file to display a message about entering hacking field.

Approach

Edit index.html file in Apache root directory and add custom text.

Commands

- `nano` → edit file
- `echo` → insert text
- `cat` → display file

Output

STEP 1 : open index file

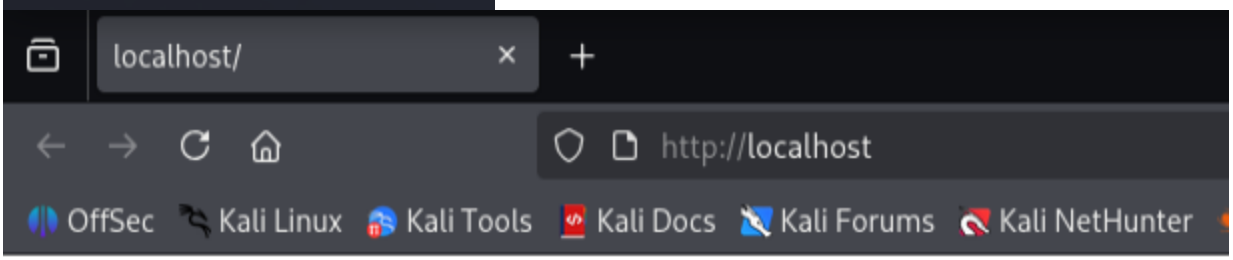
```
┌──(aman㉿kali)-[~]
└─$ sudo nano /var/www/html/index.html
```

STEP 2 : add webpage content

```
  GNU nano 8.7.1
<h1>Question 2 DONE here</h1>
```

STEP 3 : view webpage

```
┌──(aman㉿kali)-[~]
└─$ firefox http://localhost
```



Question 2 DONE here

Key Observations

Apache serves files from /var/www/html directory.
index.html is default homepage file.

What I learned

Website content can be controlled by editing HTML files.

Question 3

Enable SSH service and remotely connect to kali machine from another device in same network.

Approach

Start SSH server then connect using IP address from another system.

Commands

- `systemctl` → manage services
- `ssh` → remote login
- `ip a` → find IP address

Output

STEP 1 : start ssh service

```
┌──(aman㉿kali)-[~]
└─$ sudo systemctl start ssh
```

STEP 2 : find system IP

```
┌──(aman㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:7b:ec:7b brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 84949sec preferred_lft 84949sec
    inet6 fd17:625c:f037:2:a067:4a98:ec95:7ca5/64 scope global temporary dynamic
       valid_lft 86194sec preferred_lft 14194sec
    inet6 fd17:625c:f037:2:a00:27ff:fe7b:ec7b/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86194sec preferred_lft 14194sec
    inet6 fe80::a00:27ff:fe7b:ec7b/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

STEP 3 : connect remotely

```
┌──(aman㉿kali)-[~]
└─$ ssh aman@127.0.0.1
aman@127.0.0.1's password:
Linux kali 6.18.12+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.12-1kali1 (2026-02-25) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Fri Apr  3 06:04:13 2026 from 127.0.0.1
```

Key Observations

SSH uses port 22 by default.
Remote login requires correct username and IP.

What I learned

SSH allows secure remote command execution.

Question 4

Run MySQL server, update root password, and access internal database.

Approach

Start MySQL service, login as root, change password, then switch database.

Commands

- `systemctl` → start database service
- `mysql` → database client
- `ALTER USER` → change password

Output

STEP 1 : start mysql service

```
┌──(aman㉿kali)-[~]
└─$ sudo systemctl start mysql
```

STEP 2 : login mysql

```
┌──(aman㉿kali)-[~]
└─$ sudo mysql -u root
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 31
Server version: 11.8.6-MariaDB-2 from Debian -- Please help get to 10k stars at https://github.com/MariaDB/Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

STEP 3 : change password

```
MariaDB [(none)]> ALTER USER root@localhost IDENTIFIED BY 'hackers-arise';
Query OK, 0 rows affected (0.370 sec)
```

STEP 4 : switch database

```
MariaDB [(none)]> USE mysql;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [mysql]>
```

Key Observations

MySQL root account controls database permissions.
Weak passwords create security risk.

What I learned

Database credentials must be properly secured.

Question 5

Activate PostgreSQL service and configure it for Metasploit usage.

Approach

Start PostgreSQL database and initialize it for penetration testing tools.

Commands

- `systemctl` → manage services
- `service postgresql start` → start database
- `msfdb init` →initialize metasploit database

Output

STEP 1 : start postgresql service

```
┌──(aman㉿kali)-[~]
└─$ sudo systemctl start postgresql
```

STEP 2 : initialize metasploit database

```
┌──(aman㉿kali)-[~]
└─$ sudo msfdb init
[i] Database already started
WARNING:  database "postgres" has a collation version mismatch
DETAIL:  The database was created using collation version 2.41, but the operating system provides version 2.42.
HINT:  Rebuild all objects in this database that use the default collation and run ALTER DATABASE postgres REFRESH COLLATION VERSION, or build PostgreSQL with the right library version.
[+] Creating database user 'msf'
WARNING:  database "postgres" has a collation version mismatch
DETAIL:  The database was created using collation version 2.41, but the operating system provides version 2.42.
HINT:  Rebuild all objects in this database that use the default collation and run ALTER DATABASE postgres REFRESH COLLATION VERSION, or build PostgreSQL with the right library version.
WARNING:  database "postgres" has a collation version mismatch
DETAIL:  The database was created using collation version 2.41, but the operating system provides version 2.42.
HINT:  Rebuild all objects in this database that use the default collation and run ALTER DATABASE postgres REFRESH COLLATION VERSION, or build PostgreSQL with the right library version.
[+] Creating databases 'msf'
WARNING:  database "postgres" has a collation version mismatch
DETAIL:  The database was created using collation version 2.41, but the operating system provides version 2.42.
HINT:  Rebuild all objects in this database that use the default collation and run ALTER DATABASE postgres REFRESH COLLATION VERSION, or build PostgreSQL with the right library version.
WARNING:  database "postgres" has a collation version mismatch
DETAIL:  The database was created using collation version 2.41, but the operating system provides version 2.42.
HINT:  Rebuild all objects in this database that use the default collation and run ALTER DATABASE postgres REFRESH COLLATION VERSION, or build PostgreSQL with the right library version.
[+] Creating databases 'msf_test'
WARNING:  database "postgres" has a collation version mismatch
DETAIL:  The database was created using collation version 2.41, but the operating system provides version 2.42.
HINT:  Rebuild all objects in this database that use the default collation and run ALTER DATABASE postgres REFRESH COLLATION VERSION, or build PostgreSQL with the right library version.
[+] Creating configuration file '/usr/share/metasploit-framework/config/database.yml'
[+] Creating initial database schema
```

STEP 3 : verify database connection

```
┌──(aman㉿kali)-[~]
└─$ msfconsole
Metasploit tip: View missing module options with show missing


 _____________________________________________________________________
|                                                                     |
|                       3Kom SuperHack II Logon                       |
|_____________________________________________________________________|
|                                                                     |
|                                                                     |
|                                                                     |
|                User Name:          [   security    ]                |
|                                                                     |
|                Password:           [               ]                |
|                                                                     |
|                                                                     |
|                                                                     |
|                                  [ OK ]                             |
|_____________________________________________________________________|
|                                                                     |
|                                                https://metasploit.com |
|_____________________________________________________________________|


       =[ metasploit v6.4.112-dev                         ]
+ -- --=[ 2,607 exploits - 1,325 auxiliary - 1,707 payloads     ]
+ -- --=[ 429 post - 49 encoders - 14 nops - 9 evasion          ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf >
```

Key Observations

Metasploit stores can results in PostgreSQL database.
Database improves penetration testing workflow.

What I learned

Security tools often rely on databases to store findings.

Security Insights

1. Running web servers increases exposed attack surface.
2. Web page content can reveal attacker presence if misconfigured.
3. SSH allows remote control, making brute-force attacks possible.
4. Weak database credentials can lead to privilege escalation.
5. Metasploit databases store sensitive penetration testing data.